

“Lifestyle and Demographic Predictors of Diabetes: A Population-Level Analysis Using BRFSS Health Indicators”

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Abstract/Intro/Motivation

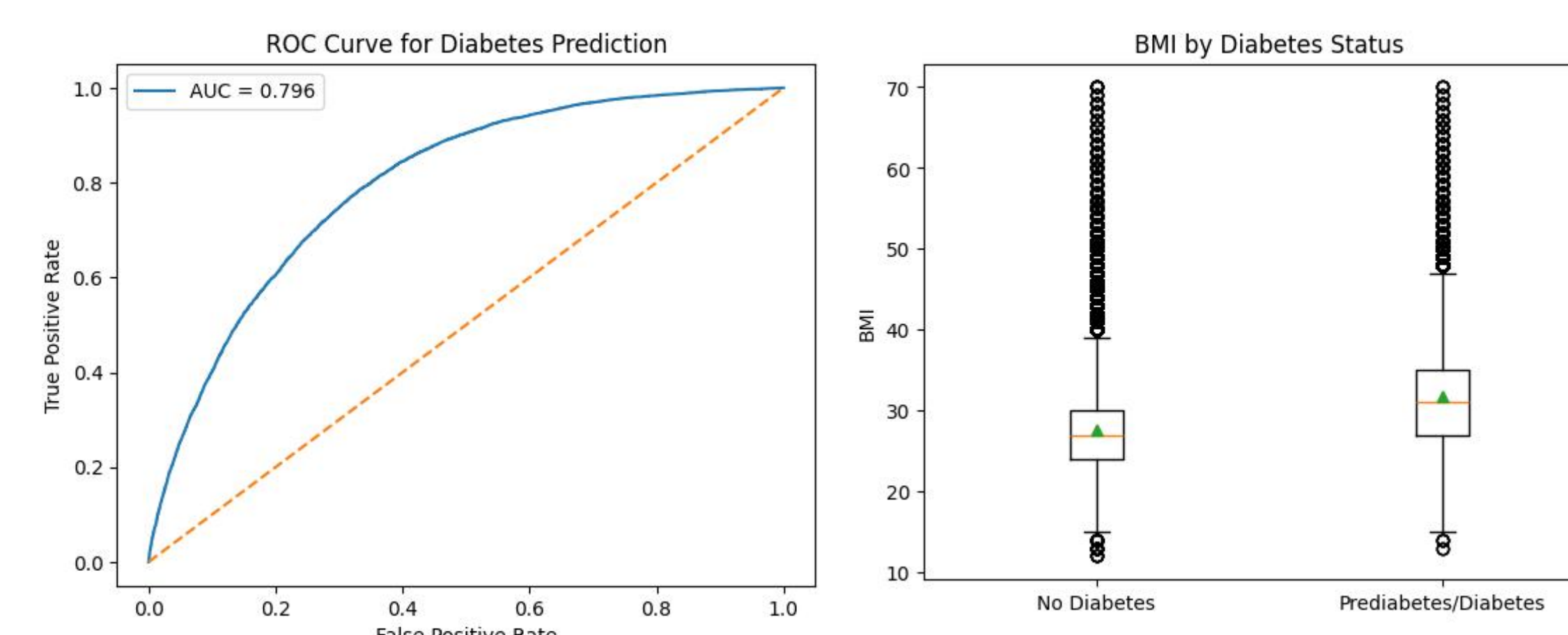
This project examines which lifestyle and demographic factors are most strongly associated with diabetes and whether they can help predict diabetes risk. Prior research — including national surveillance studies such as BRFSS — has shown that BMI, blood pressure, cholesterol, and physical activity are major contributors to diabetes. Our work builds on these findings by analyzing updated BRFSS data to evaluate these relationships in a large U.S. adult sample.

Methods/Approach

Our research used the large BRFSS Diabetes Health Indicators dataset to study how lifestyle and demographic factors relate to diabetes and to test a simple model for predicting diabetes risk. This approach is different because it applies population-level data rather than small clinical samples, offering a broader view of risk patterns. The analysis is supported by BRFSS survey data, regression models, and visuals such as BMI boxplots, correlation maps, and ROC curves.

Results

This analysis found that higher BMI, high blood pressure, and high cholesterol were the strongest factors associated with diabetes in this population, while physical activity and smoking showed weaker or inconsistent influence. Through logistic regression, we learned that even a small set of widely available health indicators can meaningfully signal diabetes risk. The prediction model performed moderately well, with an ROC curve showing reasonable discrimination between diabetic and non-diabetic individuals. The BMI boxplot clearly illustrated that individuals with diabetes tended to have higher BMI values, and the correlation matrix highlighted expected relationships among cardiovascular risk factors. These findings suggest that weight status and cardiovascular health remain central to understanding diabetes risk, and reinforce the value of simple screening measures for early identification. Overall, our results conclude that although basic health indicators can help predict diabetes, additional detailed clinical or behavioral information would likely improve prediction accuracy.



Conclusions/Future Directions

Our analysis showed that higher BMI, high blood pressure, and high cholesterol were the strongest predictors of diabetes, while physical activity and smoking played smaller roles. The logistic regression model and group comparisons demonstrated that simple, commonly collected health indicators can still offer meaningful insight into diabetes risk. The ROC curve showed moderate predictive ability, and the BMI boxplot highlighted consistently higher BMI among individuals with diabetes. These results suggest that basic risk factors remain useful for early screening, but the limits of self-reported data and the absence of clinical measures—such as HbA1c or glucose levels—reduce overall accuracy. Future work could incorporate richer clinical data, explore more advanced modeling methods, and examine differences across demographic groups to better understand and predict diabetes risk.